Quiz 3 Master Preparation Pack (Lectures 8 to 11)

Course: Environmental Science and Technology (ES1.101) • **Institute:** IIIT Hyderabad

Syllabus Scope: Lecture 8 (Tipping Points & Planetary Risks) • Lecture 9 (Carbon Budget & Terrestrial Sinks) • Lecture 10 (Kyoto Protocol, CERs & Ocean Sinks) • Lecture 11 (Negative Emission Technologies, DAC & Abatement)

Contents: Complete Past Papers Question Bank (2018–2024), 2 Timed Mock Exams with Answer Keys, and Active-Recall Checklist.

🎯 High-Frequency Quantitative Constants Quick-Check

These exact numerical values and constants are repeatedly tested across past IIIT-H exam papers:

Metric / Concept	Exact Value Expected	Syllabus Reference
Wood Biomass to Elemental Carbon	**50% (0.5 kg C per 1 kg dry wood)**. 2 tonnes biomass = 1 tonne C.	Lecture 09 (Forest Stocks)
Soil vs. Terrestrial Vegetation Carbon	**3 to 4 times** (~1,500–2,400 Gt C in soils vs ~500 Gt C in vegetation).	Lecture 09 (Soil Reservoirs)
Tundra & Boreal Soil Carbon Share	**~33% to 40%** of global soil carbon pool.	Lecture 09 (Permafrost Sinks)
Annual Young Forest CO2 Sequestration	**2 to 5 tonnes of CO_2 per hectare per year**.	Lecture 09 (Afforestation)
1 CER (Certified Emission Reduction)	**1 metric tonne of CO_2-equivalent** avoided or sequestered.	Lecture 10 (Kyoto Mechanisms)
CDM Ineligible Project Type	**Nuclear energy/power projects** (Marrakech Accords ban).	Lecture 10 (CDM Rules)
Global Warming Potential of SF_6	**23,900** (highest industrial gas GWP).	Lecture 10 (Greenhouse Gases)
Global Warming Potential of N_2O	**265 to 310** (AR5: 265; AR4: 310).	Lecture 10 (Greenhouse Gases)
Global Warming Potential of CH_4	**21 to 28** (SAR: 21; AR5: 28).	Lecture 10 (Greenhouse Gases)
Hypercane Formation Ocean Temp	**Approximately 50 °C** (winds $> 800 extkm/h$; penetrates stratosphere).	Lecture 08 (Planetary Risks)
Doomsday Clock Current Setting	**90 seconds to midnight** (driven by nuclear risk & climate change).	Lecture 08 (Existential Risks)
Biochar Pyrolysis Sequestration Potential	**Up to 4.8 Gt CO_2/year** (soil half-life $> 1000 extyears$).	Lecture 11 (NETs)

Part I: Complete Extracted Past Papers Question Bank

Extracted from 14 official past examination papers (2018–2024), organized strictly by lecture with complete model solutions.

Lecture 8: Tipping Points, Catastrophic Risks & Feedback Anomalies

FILL IN THE BLANK EVS End Sem 2022

1. Hypercane Sea Surface Temperature Threshold

"*Hypercane is a hypothetical class of extreme tropical cyclone that could form if ocean temperatures reach approximately ____ °C.*"

Model Answer: 50 °C. A hypercane would generate wind speeds exceeding $800extkm/h$ and central pressures below $650exthPa$.

FILL IN THE BLANK EST Mid 1 2024

2. Primary Factors Influencing the Doomsday Clock

"*_____ and _____ are two major factors influencing the Doomsday Clock.*"

Model Answer: Nuclear risk / nuclear weapons proliferation and **Climate change** (along with disruptive technologies). Maintained at 90 seconds to midnight by the Bulletin of the Atomic Scientists.

MECHANISMS EST Mid 1 2024, EVS Mid 2 2018

3. Why do volcanic eruptions become more frequent when glaciers melt?

Model Answer: Continental ice sheets exert immense weight compressing the Earth's lithosphere. As glaciers melt rapidly, this mass is removed (isostatic rebound). The resulting lithospheric depressurization lowers the melting point of mantle rock and allows trapped magma chambers to expand and gas bubbles to nucleate, sharply accelerating explosive volcanic eruption frequencies.

MECHANISMS EVS End Sem 2023, EST Mid 1 2024

4. How do Hypercanes differ from present-day hurricanes, and how do they impact Earth's Ozone layer?

Model Answer:

- **Difference:** Normal hurricanes require water $\geq 26.5^{\circ}extC$, reach $\sim 300extkm/h$, and are confined to the troposphere ($\sim 15extkm$). Hypercanes require ocean waters reaching $\sim 50^{\circ}extC$, reach extreme winds $> 800extkm/h$, and have vertical storm columns penetrating $40ext–50extkm$ into the stratosphere.
- **Ozone Impact:** The column lofts colossal amounts of water vapor and sea-salt aerosol chlorine radicals directly into the stratosphere, which catalytically break down stratospheric ozone (O_3), destroying Earth's UV-protective layer.

DEFINITIONS EST Mid 1 2024, EVS Mid 2019

5. Define: (a) Tipping Point, (b) Gray Goo, (c) Global Terrestrial Stilling, (d) Nuclear Winter

Model Answer:

- **Tipping Point:** A critical threshold where a tiny additional perturbation triggers a self-reinforcing, non-linear, and irreversible regime shift in the Earth system.
- **Gray Goo:** An existential disaster in molecular nanotechnology where synthetic self-replicating nanobots multiply out of control, consuming all planetary carbon into copies of themselves.
- **Global Terrestrial Stilling:** A multi-decadal decrease in near-surface wind speeds (~0.3 m/s drop per decade) caused by increased land surface roughness (urbanization, afforestation) and reduced equator-to-pole temperature gradients.
- **Nuclear Winter:** Global cooling caused by firestorms after nuclear detonations; black carbon smoke is lofted into the stratosphere, blocking sunlight for months, freezing continents, and halting photosynthesis.

MECHANISMS EVS End Sem 2023

6. Why is melting of Arctic summer ice considered the single greatest planetary threat?

Model Answer: Melting white sea ice (albedo 0.6–0.85) exposes dark ocean water (albedo 0.06), absorbing $> 90\%$ of solar radiation. This ice-albedo positive feedback causes polar amplification, disrupting the polar jet stream, destabilizing Greenland glaciers, and triggering subsea permafrost thaw that risks venting massive subterranean methane hydrates.

Lecture 9: Carbon Budget Partitioning, Forests & Soil Sequestration

FILL IN THE BLANK EVS Quiz 3 2020, Mid 2 2018, End 2019, End 2022

1. Elemental Carbon Content of Tree Biomass

"One kilogram of dry wood contains ____ kg of carbon" / "If the dry biomass of a tree is 2 tonnes, it contains ____ tonne of carbon."

Model Answer: 0.5 kg (or **50%**). A 2-tonne dry tree holds exactly **1 tonne** of carbon.

FILL IN THE BLANK EVS Quiz 3 2020, Mid 2 2018, End 2019

2. Carbon Saturation in Climax Mature Forests

"In mature forests, all gross primary productivity is either used up in ____ or returned to soil as ____ with no net addition to standing biomass."

Model Answer: Plant / autotrophic respiration; **litterfall / dead organic debris**. In mature forests, photosynthetic gains equal plant respiration plus heterotrophic litter decomposition ($GPP □pprox R_{ ext{auto}} + R_{ ext{hetero}}$), reaching carbon saturation.

FILL IN THE BLANK EVS Quiz 3 2020, Mid 2 2018

3. Soil vs. Terrestrial Vegetation Carbon Reservoir Ratio

"Soils store about ____ times as much carbon as does living terrestrial vegetation."

Model Answer: 3 times (approx. 1,500–2,400 Gt C in soil vs. ~500 Gt C in living vegetation).

CORE CONCEPTS EVS Quiz 3 2020, End Sem 2019, Quiz 4 2020

4. Differentiate between Afforestation and Reforestation

Model Answer:

- **Afforestation:** Planting trees on land that historically has *not* contained forest cover for at least 50 years under UNFCCC rules (e.g. barren lands, degraded drylands).
- **Reforestation:** Re-establishing trees on land that historically supported forests but was recently converted or cleared through deforestation, logging, or fire.

APPLIED ECOLOGY EVS Quiz 3 2020, End Sem 2019

5. Why is Agroforestry an attractive option for carbon mitigation?

Model Answer:

1. **Deep Subsoil Storage:** Tree root networks deposit carbon into deep subsoil horizons resistant to oxidation and tilling disturbance.
2. **Food Security Synergy:** Combines trees with crops/livestock without removing fertile agricultural land from food production.
3. **Fertilizer & N_2O Reduction:** Leguminous trees fix atmospheric nitrogen, enhancing soil fertility while lowering synthetic nitrogen fertilizer requirements.

CASE STUDIES EVS Quiz 3 2020, End Sem 2019

6. Describe any one International Carbon Sequestration Project

Model Answer:

- **INFAPRO Project (Sabah, Malaysia):** Partnership between Innoprise Corp and the Dutch FACE Foundation. Rehabilitated 25,000 hectares of logged, degraded Dipterocarp rainforest through enrichment planting with native wildings, sequestering millions of tonnes of carbon over a 60-year concession while restoring endangered wildlife habitats.
- *(Alternative)* **Plan Vivo / Scolel'te Project (Chiapas, Mexico):** Community agroforestry initiative where Mayan farmers manage native timber (cedar, mahogany) and shade-grown coffee, selling verified carbon credits to support rural livelihoods.

Lecture 10: Kyoto Architecture, CERs, Ocean Acidification & Geoengineering

FILL IN THE BLANK EVS Quiz 3 2020, Mid 2 2018, End 2022

1. Projects Explicitly Ineligible under CDM

"____ *projects are not eligible under CDM.*"

Model Answer: Nuclear (or Nuclear energy/power projects). Explicitly excluded under the Marrakech Accords due to proliferation hazards, safety, and nuclear waste storage concerns.

COMPARATIVE ANALYSIS EVS Quiz 3 2020, Mid 2 2018, End 2023

2. Differentiate between Annex I and Annex II Countries

Model Answer:

- **Annex I:** Industrialized OECD countries plus Economies in Transition (EITs like Russia and Eastern Europe) that committed to binding emission reduction caps under Kyoto.
- **Annex II:** A subset of Annex I (OECD members only, excluding EITs) legally mandated to provide financial resources and technology transfer to Non-Annex I developing nations.

FLEXIBLE MECHANISMS EVS End Sem 2023, Mid 2 2018, End 2022

3. Explain the Market-Based Flexible Mechanisms under the Kyoto Protocol

Model Answer:

1. **Clean Development Mechanism (CDM):** Annex I countries invest in green emission-reduction projects in Non-Annex I developing nations, generating **Certified Emission Reductions (CERs)** ($1 ext{CER} = 1 ext{t} CO_2 ext{e}$).
2. **Joint Implementation (JI):** Annex I countries invest in emission-reduction projects in other Annex I countries (mainly EITs), earning **Emission Reduction Units (ERUs)**.
3. **International Emissions Trading (IET):** Annex I nations trade assigned surplus allowance units (AAUs) directly among themselves.

MARINE CHEMISTRY EVS Quiz 3 2020, End Sem 2023

4. Ocean Acidification & Logarithmic Sensitivity

"What are the effects of ocean acidification, and why be concerned with a small change in ocean pH?"

Model Answer: Dissolved CO_2 produces carbonic acid ($CO_2 + H_2Oightleftharpoons H_2CO_3ightleftharpoons H^+ + HCO_3^-$). Surplus H^+ reacts with carbonate: $H^+ + CO_3^{2-}ightleftharpoons HCO_3^-$.
Because pH is a logarithmic scale ($-\log_{10}[H^+]$), a 0.1 unit drop represents a **~30% increase in** $[H^+]$. Consuming carbonate drops the saturation state ($\Omega < 1$) of aragonite and calcite, dissolving shells of corals, pteropods, and mollusks.

GEOENGINEERING HAZARDS EVS Mid 2 2018, End 2019, End 2023

5. John Martin's Ocean Iron Hypothesis & Ecological Hazards of Ocean Fertilization (Sulu Sea)

Model Answer:

- **Iron Hypothesis:** Iron is the limiting micronutrient in High-Nutrient Low-Chlorophyll (HNLC) oceans. Seeding iron triggers diatom blooms that sequester carbon to deep sediments ("*Give me half a tanker of iron, and I'll give you an ice age*").
- **Ecological Hazards (Sulu Sea):** Stimulates toxic diatoms (*Pseudo-nitzschia*) that produce lethal **domoic acid** (amnesic shellfish poisoning); sinking biomass decomposition creates deep-water anoxia; anoxic decay emits potent GHGs (N_2O and CH_4).

CLEANTECH MATERIALS EVS End 2023, End 2019, Mid 2 2018

6. Difference between Portland Cement and Novacem Cement

Model Answer:

- **Portland Cement:** Made from limestone ($CaCO_3$) calcined at $1,450^\circ extC$; releases decarbonation and fuel emissions of $0.8ext–1.0exttCO_2$ per ton cement produced.
- **Novacem Cement:** Uses magnesium silicates calcined at $650^\circ extC$. Absorbs CO_2 during curing ($MgO + CO_2oMgCO_3$), sequestering $0.6exttCO_2$ per ton produced (net carbon negative).

Lecture 11: Negative Emission Technologies, DAC & Cleantech Solutions

CORE CONCEPTS EVS End Sem 2022, End Sem 2023

1. Blue Carbon Habitat Restoration & 3 Key Advantages

Model Answer: Blue carbon is carbon stored by coastal marine ecosystems (mangroves, tidal salt marshes, seagrass meadows).
Three Key Advantages:

1. **No Saturation:** Sequestration rate is 3 to 5 times (and up to 40x per unit area) higher than terrestrial forests; sediments remain anoxic and saline, locking carbon for millennia without saturating.
2. **Coastal Protection:** Buffers shorelines by attenuating wave energy from storm surges and cyclones.
3. **Marine Habitat:** Serves as critical breeding and nursery grounds for commercial fish and crustaceans.

FILL IN THE BLANK EVS End Sem 2022

2. Livestock Enteric Methane Mitigation via Seaweed

"Methane reduction from livestock can be achieved by adding _____ to cattle feed."

Model Answer: Seaweed (specifically the red macroalga *Asparagopsis taxiformis*). Contains **bromoform** ($CHBr_3$) which inhibits methyl-coenzyme M reductase in rumen methanogenic archaea, reducing methane emissions by $80ext–90\%$.

EXOTIC CLEANTECH EVS End Sem 2022, End Sem 2023

3. Diamonds from Air Pollution (Aether Diamonds Manufacturing Process)

Model Answer:

1. **Capture:** Ambient CO_2 is captured directly from air via Direct Air Capture (Climeworks).
2. **Methanation:** CO_2 is combined with green hydrogen (H_2) via the Sabatier reaction ($CO_2 + 4H_2oCH_4 + 2H_2O$) to produce high-purity methane.
3. **Chemical Vapor Deposition (CVD):** Methane is cracked in microwave plasma CVD reactors, depositing carbon atom-by-atom to grow gem-grade diamonds.

CHEMICAL REACTION Lecture 11 Core Syllabus

4. Enhanced Silicate Rock Weathering (Olivine Reaction)

Model Answer: Spreading micro-pulverized olivine on agricultural soils reacts with soil rainwater and carbonic acid:

$$Mg_2SiO_4 + 4CO_2 + 4H_2Oo2Mg^{2+} + 4HCO_3^- + H_4SiO_4$$

Converts gaseous CO_2 into stable aqueous bicarbonate (HCO_3^-) washed to the ocean, sequestering carbon for $> 100,000extyearsextyearsextyearsext

Part II: Official Mock Examination 1 (25 Marks)

Time: 45 Minutes | **Syllabus:** Lectures 8 to 11 | **Format:** Official IIIT-H Quiz 3

Section 1: Fill in the Blanks (10 × 0.5 = 5.0 Marks)

1. _____ projects are explicitly NOT eligible for carbon credits under CDM.
2. In mature forests, all gross primary productivity is either used up in _____ or returned to soil as _____ with no net addition to standing biomass.
3. _____ is the single major source of agricultural enteric methane emissions globally.
4. _____ stores carbon while simultaneously intercepting agricultural nutrient runoff to protect water quality.
5. A growing young forest can remove approximately _____ tonnes of CO_2 per hectare per year.
6. Soils store approximately _____ times as much organic carbon as does living terrestrial vegetation.
7. One kilogram of dry wood contains _____ kg of elemental carbon.
8. Approximately _____ % of global soil carbon is locked in tundra and boreal permafrost ecosystems.
9. Nitrous Oxide (N_2O) has a Global Warming Potential (GWP) of _____ times that of CO_2.
10. Hypercanes are hypothetical super-cyclones that form when sea surface temperatures reach approximately _____ °C.

Section 2: Expand the Abbreviations (5 × 1.0 = 5.0 Marks)

1. **IPCC** | 2. **UNFCCC** | 3. **CER** | 4. **GWP** | 5. **CDM**

Section 3: Differentiate Between (2 × 2.0 = 4.0 Marks)

1. **Afforestation** vs. **Reforestation**

2. **Annex I** vs. **Annex II** countries under the UNFCCC / Kyoto Protocol

Section 4: Technical & Ecological Mechanisms (2 × 2.5 = 5.0 Marks)

1. **Ocean Acidification:** Write the chemical equilibrium reaction when CO_2 dissolves in seawater, and explain why calcifying organisms face shell dissolution even under slight pH shifts.

2. **Urea Fertilization:** Explain the mechanism of ocean nitrogen fertilization and detail two major ecological hazards caused by sinking organic matter.

Section 5 & 6: Applied Systems & Case Studies (2 × 3.0 = 6.0 Marks)

1. Define **Agroforestry** and list three scientific reasons why it represents an attractive carbon mitigation strategy.

2. Describe either the **INFAPRO Project** in Malaysia or the **Plan Vivo Project** in Mexico.

📋 Mock Exam 1: Official Answer Key & Grading Scheme

Section 1 Answers (0.5M each):

1. **Nuclear** (or Nuclear power / Nuclear energy projects).
2. **Plant / autotrophic respiration**; **litterfall / dead organic debris**.
3. **Enteric fermentation in ruminant livestock** (cattle/sheep).
4. **Riparian forest buffers** (or Riparian buffers).
5. **2 to 5 tonnes** of CO_2/ha/year.
6. **3 times** (accept 3 to 4 times; ~1500–2400 Gt in soil vs ~500 Gt in vegetation).
7. **0.5 kg** (50% carbon fraction).
8. **33% to 40%** (approx. one-third).
9. **265 to 310** (265 under IPCC AR5; 310 under AR4).
10. **50 °C**.

Section 2 Expansions (1.0M each):

1. **IPCC:** Intergovernmental Panel on Climate Change.
2. **UNFCCC:** United Nations Framework Convention on Climate Change.
3. **CER:** Certified Emission Reduction ($1extCER = 1exttCO_2exte$).
4. **GWP:** Global Warming Potential.
5. **CDM:** Clean Development Mechanism.

Section 3 Differences (2.0M each):

1. *Afforestation:* planting on land without forest for $\geq 50extyearsextyearsextyearsext

Part III: Mock Examination 2 (35 Marks — Mid-Sem Format)

Time: 60 Minutes | **Syllabus:** Lectures 8 to 11 | **Format:** High-Yield Comprehensive Exam

Section I: Rapid Technical Questions (5 × 2.0 = 10 Marks)

1. Why do explosive volcanic eruptions become significantly more frequent when continental glaciers retreat? (2M)

2. Explain the process by which a Hypercane damages the Earth's stratospheric ozone layer. (2M)

3. What is **Global Terrestrial Stilling**, and what are the two primary planetary causes? (2M)

4. State the **Sherwood Plot rule** for Direct Air Capture and explain why capturing CO_2 from ambient air consumes more thermodynamic work than flue gas capture. (2M)

5. How does the red seaweed *Asparagopsis taxiformis* suppress bovine enteric methane emissions? Name the key active compound. (2M)

Section II: Comparative Analysis (10 Marks)

1. Differentiate between the three flexible mechanisms of the Kyoto Protocol: **CDM**, **Joint Implementation (JI)**, and **International Emissions Trading (IET)**. (3M)

2. Compare conventional **Portland Cement** with **Novacem Cement** in terms of raw materials, calcination temperature, and net carbon balance. (3M)

3. Contrast **John Martin's Ocean Iron Hypothesis** with **Enhanced Silicate Rock Weathering** on agricultural lands. Give the chemical equation for olivine weathering. (4M)

Section III: Deep Technical & Case Studies (3 × 5.0 = 15 Marks)

1. **Direct Air Capture & Carbon Utilization:** Detail the 3-step technological pipeline used by **Aether Diamonds** to convert ambient air pollution into certified gem-grade diamonds. Name two leading commercial DAC companies. (5M)

2. **Planetary Tipping Cascades & Black Swan Risks:** Define Taleb's **Black Swan** concept. Describe the three major planetary tipping elements: Arctic Summer Sea Ice Loss, WAIS Collapse, and Amazon Savannization. (5M)

3. **Blue Carbon & Biochar Pyrolysis:** Define **Blue Carbon** and explain why coastal salt marshes/mangroves sequester carbon without saturation. Detail the process of **Biochar Pyrolysis** (temperature range, soil residence time, global annual potential). (5M)

📋 Mock Exam 2: Model Answers & Rubric

Section I:

1. *Glacial Isostatic Rebound:* Mass unloading causes mantle depressurization, lowering rock melting points and venting magma chambers.

2. *Hypercane Ozone Decay:* $50extkm$ vertical column injects water vapor and chlorine radicals into stratosphere, catalytically destroying O_3.

3. *Stilling:* Surface wind slowdown (~0.3 m/s drop per decade) due to surface roughness and reduced equator-to-pole thermal gradient.

4. *Sherwood Rule:* Minimum work $W_{\min} = -RT[\ln x + ↑rac{1-x}{x}\ln(1-x)]$. Atmospheric CO_2 is dilute ($420extppm = 0.042\%$), requiring $\sim 3imes$ higher theoretical energy and massive contactor fans.

5. *Seaweed:* Active compound is **bromoform** ($CHBr_3$); inhibits methyl-coenzyme M reductase in rumen archaea, cutting methane by $80ext–90\%$.

Section II:

1. *CDM:* Annex I o Non-Annex I (CERs). *JI:* Annex I o Annex I (ERUs). *IET:* Trading AAUs among Annex I nations.

2. *Portland:* Limestone ($CaCO_3$) calcined at $1450^\circ extC$; emits $+0.8ext–1.0exttCO_2/extt$. *Novacem:* Magnesium silicate calcined at $650^\circ extC$; absorbs CO_2 during curing, netting $-0.6exttCO_2/extt$.

3. *Iron Hypothesis:* Fe micronutrient in HNLC ocean stimulates diatom sink. *Enhanced Weathering:* Micro-pulverized olivine on soil: $Mg_2SiO_4 + 4CO_2 + 4H_2Oo2Mg^{2+} + 4HCO_3^- + H_4SiO_4$.

Section III:

1. *Aether Pipeline:* (1) DAC CO2 capture; (2) Sabatier methanation ($CO_2 + 4H_2oCH_4 + 2H_2O$); (3) Microwave plasma CVD diamond reactor. DAC companies: Climeworks (solid amine) & Carbon Engineering (liquid KOH).

2. *Black Swan:* Unpredictable outlier with extreme impact, rationalized in hindsight. 3 Tipping elements: Arctic ice (albedo drop), WAIS (marine ice sheet instability commits 3–5m sea level rise), Amazon (loss of biotic pump converts to scrubland, venting 100 Gt C).

3. *Blue Carbon:* Coastal wetlands build soil vertically in saline anoxic water without microbial decay. *Biochar:* $400ext–700^\circ extC$ anoxic pyrolysis; $> 1000extyr$ soil residence time; up to $4.8extGtCO_2/extyr$ potential.